# PROJECT PROPOSAL

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# Business Context

## Providing information regarding the producers

My client has asked me to provide information regarding the producers, their farming methods and the benefits of buying locally. This solves the problem of the customers not knowing where their food comes from or who produced it. By displaying clear producer profiles, buyers can learn about the origin of the products that they are purchasing, this benefits the company as it increases transparency between producers and customers. As a result the customers are way more likely to trust the platform and support local farmers

## View products with pricing and availability

My client has asked me to allow customers to view products with availability on the website. This solves the problem of customers needing to contact producers to ask about prices or whether items are in stock. By displaying this information clearly, customers can quickly decide whether they want to purchase a product. This also prevents confusion and reduces unnecessary and timewasting communication between producers and buyers, the benefit is that the purchasing becomes much more efficient and faster for both parties.

## Placing orders for Collection/Delivery

My client has asked me to implement a feature that allows customers to place an order for delivery or collection. This solves a problem of limited purchasing options for buyers who may not be available to collect the produce that they have purchased. By offering both options, the system becomes more flexible for different user needs. Customers who prefer convenience and genuinely

haven't gotten time to go and collect can choose delivery, while others can collect their orders, the benefit is that this will become more accessible to a wider range of customers.

### Account management

My client has asked me to allow customers to manage their accounts and view their order history. This solves the problem of users needing to repeatedly enter their account credentials or remember previous order purchases. By storing account information and previous orders, customers can quickly review their activity. This makes it easier to reorder products they have purchased before. The benefit is that the website provides a more personalised and organised experience for users.

### Providing a dashboard for producers

My client has asked me to provide a dashboard for producers to manage their stock levels, update produce details and view orders. This essentially solves the problem of producers having to manually track their stock and orders. With a central dashboard, all important information is available in one place. Producers can conveniently update their product information and monitor demand. The benefit is that it improves organisation and helps producers manage their products more efficiently.

### Order management, scheduling and tracking features

My client has asked me to implement order management, scheduling and tracking features within the system. This solves the problem of order becoming disorganised when managed manually, by organising orders digitally, producers and customers can clearly see the order status and delivery schedules. This improves communication between the hub, producers and customers. The benefit is that it helps the business run more smoothly and reduces operational errors.

### Discounts and special offers

My client has asked me to implement discounts and special offers on certain products. This solves the problem of attracting and retaining customers in a competitive market. By offering to purchase products they may not normally buy. This can also help producers sell surplus or seasonal produce more quickly. The benefit is that it increases sales and encourages repeat purchases.

### Accessibility features to support a wide range of users

My client has asked me to implement accessibility features such as dark mode and adjustable display settings. This solves the problem of some users struggling to comfortably use the website due to visual or usability issues. By allowing users to customise how the website appears, the platform becomes easier to navigate for more people. This ensures the system supports a wider range of users. The benefit is that the website becomes more inclusive and user friendly.

## Project Aim

The overall aim of this project is to provide Greenfield Local Hub with a functional website that:

- Provides information regarding the producers
- Viewing products with pricing and avaliability
- Placing orders for collection or delivery
- Manage their account and viewing order history
- Providing a dashboard to enable producers to manage stock levels, update product details and view orders.
- Account management, scheduling and tracking features

- Discounts and special offers
- Accessibility features to support a wide range of users.

# Functional Requirements and User Acceptance Criteria

## Users Personas

Franklin – A Local customer buying fresh produce

Michael – A local farmer/producer

Trevor – A customer with accessibility needs for comfort

## Functional Requirements Table

| **Functional Requirement** | **User Story** | **User Acceptance Criteria** | **Justification** |
|---|---|---|---|
| Users being able to register | As Franklin, Michael and Trevor, I would want to be able to register so I'm able to access features on the site | • View register option on the right side of the navigation menu<br>• Click on the register button<br>• Enter all the required details, such as name password and email<br>• Submit the form to create an account<br>• Receive a confirmation and be redirected to the log in page<br>• The page must support and comply with the WCAG guidelines and the must support accessibility options such as adjustable | Account creation will allow users to securely access personalized services such as placing orders, managing products or tracking activity. This also allows the system to store user data for account management and order processing |

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| | | text size, high contract and dark/light mode. | |
| Users being able to log in | As Franklin, Michael and Trevor, I want to log into my account so I can access my very own personal dashboard and features | • View Login button in the navigation bar<br>• Selecting login to open the login page<br>• Enter registered email and password<br>• Submit login form<br>• System verifies your details and rediretts user to their dashboard<br>• The page must support and comply with the WCAG guidelines and the must support accessibility options such as adjustable text size, high contract and dark/light mode. | This feature enables returning users to access their saved information without needing to create a new account eachtime, improving usability and efficiency. |
| Users being able to change/update their login details | As Franklin, Michael or Trevor, I want to update my account details so my personal information remains accurate and unforgettable | • Log into account<br>• Access account settings page<br>• View editable profile details such as name, email and password<br>• Modify required information | Allowing users to update their account details ensures personal data is accurate and helps maintain secure login credentials |

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| | | • Confirm changes with a save button<br>• Updated info is stored successfully<br>• The page must support and comply with the WCAG guidelines and the must support accessibility options such as adjustable text size, high contract and dark/light mode. | |
| Users able to manage their accounts and view order history | As Franklin or Trevor, I want to be able to manage my account and view my order history, so I can keep track of my purchases and reorder products easily | • Log in or register to access account features<br>• View account button in the navigation bar<br>• Interact with account button to navigate to account dashboard<br>• View personal details and account settings<br>• View order history section displaying previous orders<br>• Select previous orders to view full details<br>• The page must support and comply with the WCAG | This solves the problem of customers forgetting previous purchases or repeatedly entering their details. Allowing users to manage their account and view order history makes the platform more organised and encourages repeat purchases. |

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| | | guidelines and the must support accessibility options such as adjustable text size, high contract and dark/light mode. | |
| Users being able to view information regarding the producers | As franklin or trevor, I want to view information about the producers so that I can learn where the products came from and how they are essentially produced. | • View producers section on the navigation bar<br>• Interact with producers with images and brief descriptions<br>• Select producer profile to view more detailed information such as farming methods and background<br>• The page must support and comply with the WCAG guidelines and the must support accessibility options such as adjustable text size, high contract and dark/light mode. | This solves the problem of customers not knowing who produces the food that they are buying. Providing information about the producers increases transparency and helps build trust between customers and local farmers. |
| Customers able to place orders for collection or delivery | As Franklin or Trevor, I wan to place an order and select either collection or delivery so that I can receive my produce in a way that suits me | • Login or register to access ordering features<br>• Browse products and select items | This solves the problem of limited purchasing methods for customers. Offering both collection and delivery provides flexibility and allows the business to |

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|---|---|---|---|
| | | • Click add to basket button on product page<br>• Navigate to checkout page<br>• Select collection or delivery option<br>• Enter delivery details if delivery option is selected<br>• Confirm order and receive confirmation<br>• The page must support and comply with the WCAG guidelines and the must support accessibility options such as adjustable text size, high contract and dark/light mode. | reach more customers. |
| Producers able to access a dashboard to manage products and stock | As Michael, I want to access a producer dashboard so that I can manage my stock levels and update product details and view orders | • Log in or register<br>• Navigate to producer dashboard<br>• View list of products that are added by the producer<br>• Edit product information such as description, prices and images<br>• Update stock quantities<br>• View list of orders | This solves the problem of producers needing to manage their products and orders manually. A dashboard allows them to update product information and monitor stock levels efficiently. |

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| | | containing the producers products<br>• The page must support and comply with the WCAG guidelines and the must support accessibility options such as adjustable text size, high contract and dark/light mode. | |
| Users able to view products with pricing and availability | As franklin or Trevor, I want to view products with their prices and availability so that I can decide on what I want to purchase | • View products section on the homepage<br>• View product cards displaying product name, price and availability status<br>• Select product card to view full product details<br>• View information such as price, description and stock availability<br>• The page must support and comply with the WCAG guidelines and the must support accessibility options such as adjustable text size, high contract and dark/light mode. | This solves the problem of customers needing to contact producers to check prices or availability. Displaying this information clearly allows customers to make faster purchasing decisions. |

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| Systems able to manage orders, scheduling and tracking | As franklin, or trevor, I want to track my order status so that I know when my order will be ready or delivered. | • Log in to account<br>• Navigate to orders section<br>• View list of current orders<br>• View order status such as processing, scheduled, ready for collection or delivered<br>• View estimated collection or delivery time<br>• The page must support and comply with the WCAG guidelines and the must support accessibility options such as adjustable text size, high contract and dark/light mode. | This solves the problem of customers not knowing what stage their order is at. Tracking and scheduling features improves organisation and communication between producers and the cistomers |
| Producers being able to apply discounts and special offers | As Michael, I want to apply discounts or special offers to products so that I can encourage customers to buy more produce. | • Log in to producer dashboard<br>• Navigate to product management section<br>• Select product to edit<br>• Enter discount percentage or promotional price<br>• Save changes<br>• Discount price appears on product listing<br>• The page must support and comply with | This solves the problem of producers struggling to promote certain products or seasonal items. Discounts and promotions encourage more purchases and help increase sales. |

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| | | the WCAG guidelines and the must support accessibility options such as adjustable text size, high contract and dark/light mode. | |
| Websites are able to provide accessibility features for users | As Trevor, I want accessibility options such as dark mode and adjustable text size so that I can comfortably use the website | • View accessibility settings option on the website<br>• Select display options such as light mode or dark mode<br>• Adjust text size if necessary<br>• Websites also updates display settings instantly<br>• User can continue browsing with accessibility settings applied. | This solves the problem of some users finding websites difficult to use due to visual or usability limitations. Accessibility features ensure the website can be used by a wider range of people. |

# Non-Functional Requirements

## Performance

| **Requirement** | **Justification** | **KPI** |
|---|---|---|
| The system should remain stable even when usage increases beyond normal expectations | This ensures he website can still function properly during unusually busy periods without crashing or becoming unusable, this helps prevent loss of potential customers | The system should continue operating correctly with up to 4 times the expected load |
| CPU consumption must stay within a reasonable limit while the system is running | Keeping CPU usage low allows the website to operate efficiently on both high-end and lower-powered devices without reducing performance | CPU usage should stay below 18% on an iphone 12 and below 12% on a laptop with 16gb ram |

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| Online payments must be completed within a reasonable processing time | Customers expect transactions to be quick and smooth. Slow payments could lead to frustration or buyers even changing their minds | Payment confirmation should occur within standard payment processing times used by e-commerce platforms. |
| The platform should support multiple simultaneous interactions without a reduction in performance | If many people are using the website at the same time, the core features must still operate effectively to ensure reliability. | The system should manage around 30 requests per second without slowing down. |
| Pages within the website should appear quickly after being requested by the user | Fast page rendering improves usability and reduces frustration, helping the website maintain a positive reputation with visitors | Each webpage should load fully within 15millisecondsa |

## Security

| **Requirement** | **Justification** | **KPI** |
|---|---|---|
| Passwords must meet strong security requirements before a user can create an account | Enforcing stronger passwords makes it significantly harder for attackers to gain access through password guessing or automated attacks | Passwords must contain at least 8-10 characters, including letters, numbers and a special character |
| The system must request permission before cookies are stored on a users device | Asking for consent ensures the website complies with the privacy regulations such as GDPR, and gives users control over how their data is being used. | A cookie consent banner must appear on the first visit allowing users to accept or even reject cookies. |
| Personal account info shouldn't be fully visible on the screen | Masking personal data prevents sensitive information from being leaked by somebody who happened to be nearby while you was accessing them | Sensitive fields such as emails and passwords should be displayed as asterisks only, keeping it safe from anybody who happens to take a peek |
| Users must only be able to access features relevant to their account role | Restricting features based on user roles prevents unauthorised access to sensitive system functions | Role-based permissions should ensure administrative tools are only accessible to authorised admin accounts, in this case it is producers. |
| The card holders full name shouldn't be completely visible during payment confirmation | Reducing the amount of visible personal information protects the users identity and improves privacy | Only the surname of the cardholder should appear during confirmation screens |
| Access to personal accounts must require authentication | Requiring login credentials ensure that only authorised users can view or modify their account details | Users must log in with valid credentials before accessing account pages or personal info |

## Capacity

| Requirement | Justification | KPI |
|---|---|---|
| The platform should be capable of handling increased usage as the business grows | Planning for future growth ensures the system remains usable without needed any complete redesign | The infrastructure should allow scaling to double the normal number of users and stored data |
| File storage must support uploaded content such as images without reaching limits quickly | Product images and user uploads require storage space so the system must support growth without frequent upgrades | The system should support at least 20gb of media storage for uploaded images and files |
| The system must allow a large number of orders to e processed overtime | Ensuring the platform can handle a high number of transactions prevents system limitations as the business grows | The system should support at least 2,000 orders per day without any errors or data loss |
| The platform must allow multiple users to access the website at the same time without degradation or crashing | Allowing many users ensures the site remains accessible during busy periods such as promotions or peak hours | The system should support at least 500-1000 accessing the site simultaneously |
| The system should support a large number of registered users without affecting performance | As the website grows, more users will create accounts. Ensuring the system can store and manage many users prevent limitations in further expansion | The system should support at least 10,000 registered user accounts without performance issues |

## Scalability

| Requirement | Justification | KPI |
|---|---|---|
| The system architecture should allow resources to be increased as the number of users grow | As more users begin using the website, the system must be able to expand without requiring a complete redesign | The system should support an increase of at least 100% in active users without requiring any major structural changes |
| The database should be able to handle increasing volumes of data | Overtime the platform will store more user accounts, products, and orders and so the database must scale efficiently to maintain performance | The database should maintain normal performances when data volume increases by up to 2 times its original size |
| The system should support adding additional servers or resources when demand increases | Allowing additional infrastructure to be added ensures the website can continue operating smoothly during growth or high demand periods | The system should allow additional server resources to be integrated without down time |
| System performance should remain stable as the number of users grows | Maintaining performance during growth prevents slower loading times or service interruptions that could | Response times shouldn't increase by more than 20% when the number of users doubles |

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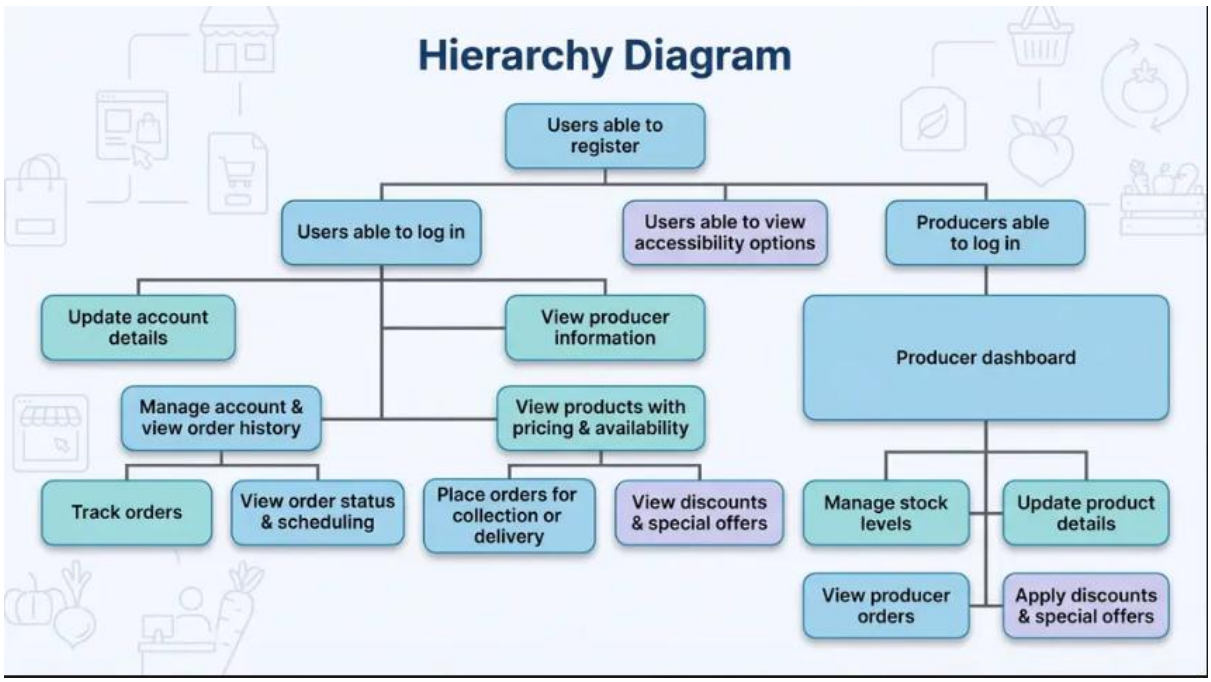
| | negatively affect user experience | |
| New functions should be implemented at an industry standard pace to the site | This will lead to the client to show that they care about their customers user experience using the site, which will make the client seem more trustworthy | On average, every new major function should be added within 5-10 days |

## Usability and Accessibility

| **Requirement** | **Justification** | **KPI** |
|---|---|---|
| The website interface should be simple and easy to understand for new users | A clear and intuitive interface helps users quickly learn how to use the platform without confusion or frustration | At least 80% of test users should be able to complete key tasks (such as browsing products or placing an order) without assistance |
| Navigation throughout the website should be clear and consistent | Consistent navigation allows users to easily move between pages and locate features without getting lost | All pages should include a visible navigation menu allowing users to access main sections of the website within three clicks or fewer |
| Text displayed on the website should be easy to read | Clear text improves readability and ensures users can understand he info being displayed without difficulty | The website should remain readable font sizes, such as 16px and maintain sufficient colour contrast between text and background |
| Forms and input fields should clearly explain the information required. | Providing clear instructions reduces user mistakes and improves the ease of completing actions such as creating accounts or placing orders | Form fields must include labels and validation messages to guide users when entering incomplete information |
| Important website functions should be accessible using a keyboard | Some users cannot rely on a mouse, so keyboard navigation ensures the website is usable for people with accessibility needs | Users should be able to navigate core features using keyboard inputs such as tab and enter |

## Hierarchy Diagram




## User Interface (UI) Considerations

During the design stage of the website, several visual and usability elements have been carefully considered to ensure the platform is easy to use and visually consistent with the purpose of the greenfield local hub. One of the main design considerations I will be making is the overall colour palette used across the platform. The website will use natural tones such as green, beige, and soft earth colours to reflect the theme of agriculture, fresh produce, and environmental sustainability. These colours will help in creating a visual connection between the website and the local farms that supply the products.

Another important factor is the layout and organisation of content on each page. The interface will avoid overcrowding by using sufficient spacing between sections and elements. This improves readability and ensures that users can easily locate important features such as product listings, navigation menus, and checkout options. A Structured layout will also make the website look more professional and much easier to navigate.

Visual elements such as images will also be used to support the design of the site. For example, photographs of farms, fresh produce, and local farmers will be displayed on relevant pages. These images help communicate the authenticity of the products and strengthen the relationship between customers and producers. Using high quality images throughout the platform will also improve the overall appearance and professionalism of the website

Typography is also an important part of the design as the site will make use of Montserrat typeface for headings and main content because it provides a clear, modern appearance. This font is commonly used across many online platforms, which means most users will be somewhat familiar

with its style and readability. Maintaining the same font throughout the platform will ensure consistency across all pages.

In addition to visual design, accessibility has also been considered in the interface design. The system will include options that allow users to adjust aspects of the display according to their needs. Examples include dark mode, adjustable font sizes, and improved colour contrast settings. These features ensure that users with different visual preferences or accessibility requirements can still comfortably interact with the website.

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## Security Risks and Mitigation Strategies

**Risk:** An attacker may attempt to access user accounts by repeatedly attempting different password combinations in order to guess a valid login.

**Mitigation:** The system will enforce strong password rules to ensure that users create secure credentials. Passwords will require a combination of letters, numbers, and special characters. Additionally, the system will temporarily restrict login attempts after multiple failed attempts to reduce the risk of automated password guessing.

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**Risk:** A malicious user may attempt to inject harmful database commands into input fields in order to manipulate or retrieve sensitive information from the system.

**Mitigation:** The application will validate and sanitise all user input before processing it. In addition, the system will rely on **parameterised database queries** which prevent database commands from being executed if they are entered through form fields.

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**Risk:** A user could attempt to insert harmful scripts into form fields such as reviews or comments that may execute when another user views the page.

**Mitigation:** All content submitted by users will be filtered and encoded before being displayed. This ensures that any potentially harmful code will be treated as plain text rather than being executed within the browser.

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**Risk:** The website could experience unusually high numbers of automated requests which may reduce performance or temporarily disrupt access for genuine users.

**Mitigation:** The system will require users to be authenticated before accessing key features such as placing orders or managing products. This reduces the ability of automated bots to misuse the system. Additional request limits may also be applied to restrict excessive activity within short time periods.

## Legal and Regulatory Considerations

Because the platform will collect personal information from its users, it must comply with relevant data protection regulations. The website will therefore follow the requirements of UK GDPR and the Data Protection Act 2018**.** These regulations ensure that organisations handle personal data responsibly and provide users with control over how their information is used.

When users register for an account or place an order, personal information such as names, email addresses, and order details will be stored by the system. To protect this information, passwords will be securely encrypted and all data transfers between the website and the user will be protected using HTTPS encryption**.**

The website will also provide users with a privacy policy page. This document will clearly explain what information is collected, why it is required, and how it is stored or used by the platform. It will also inform users about their rights regarding their personal information.

Since the platform will use cookies to improve functionality, such as remembering login sessions or user preferences, it must also obtain permission from users before storing cookies on their devices. To achieve this, a cookie consent notice will appear when a user first visits the site, allowing them to either accept or reject the use of cookies.

Accessibility regulations are also an important consideration for the system. The website will aim to follow the Web Content Accessibility Guidelines (WCAG) to ensure it can be used by individuals with different needs or disabilities. This includes features such as clear navigation, readable text sizes, and compatibility with keyboard navigation.

Finally, the platform will provide a Terms and Conditions page outlining the rules for using the service. This document will explain acceptable user behaviour, outline the responsibilities of the platform, and help protect the organisation from legal disputes related to misuse of the website.